SMARTCART

E-commerce Customer Segmentation System

Customer Segmentation Analysis & Marketing Strategy Report

PROJECT PURPOSE

Apply customer clustering to uncover groups with similar income, demographics, purchasing behaviour, shopping-channel preferences, and campaign responses. Translate the results into practical marketing and customer-retention strategies.

4	2,240	32.01%	$1,237
Customer segments	Customer records	Highest campaign response rate	Highest average spending

A business-focused interpretation of the four customer clusters and their reported results.

Executive summary

The SmartCart customer segmentation analysis identifies four distinct customer groups from a dataset of 2,240 e-commerce customer records. The groups differ in average income, spending, household composition, website activity, shopping-channel behaviour, and campaign response.

The key result is that **Cluster 3 has the highest campaign response rate (32.01%)**, while **Cluster 1 has the highest average spending ($1,237)**. Clusters 0 and 2 record more monthly website visits but much lower average spending, suggesting an opportunity to improve conversion from browsing to purchasing.

The findings can guide targeted campaigns, loyalty initiatives, and customer-retention activities. They are descriptive results from the current clusters; recommended actions should be tested and measured rather than treated as guaranteed outcomes.

1. Project overview

SmartCart is a growing e-commerce platform that wants to move beyond generic marketing. This project uses customer segmentation (clustering), an unsupervised machine learning approach, to group customers according to similarities in their data.

The analysis considers customer income and demographics alongside spending across product categories, purchase frequency, preferred shopping channels, website visits, and campaign responses. The business objective is to understand customer groups and develop more relevant strategies to improve engagement, sales, loyalty, and marketing return on investment (ROI).

2. Cluster analysis and findings

Cluster 0 – Family-oriented, lower-spending customers

Customer profile: Lower income, a higher average number of children, and typically living with a partner.

Reported results: Average income: **$39,681** | Average spending: **$222** | Average children: **1.24** | Website visits: **6.31/month** | Campaign response: **7.62%** | Store purchases: **4.14**

Interpretation: Customers in this segment generally have lower incomes and more children. They visit the website relatively frequently, but their average spending is low and their campaign response rate is the lowest of the four clusters.

Recommended strategies:

- Offer family bundles, household-product discounts, and value-for-money options.
- Introduce budget-friendly products and seasonal promotions.
- Test coupons and incentives designed to encourage repeat purchases.
- Use personalized email campaigns featuring practical products and clear savings.

Cluster 1 – Higher-value, partner-oriented customers

Customer profile: Higher income, fewer children on average, and typically living with a partner.

Reported results: Average income: **$72,808** | Average spending: **$1,237** | Average children: **0.51** | Catalog purchases: **5.50** | Store purchases: **8.66** | Campaign response: **16.67%**

Interpretation: This segment has the highest average spending and strong purchasing activity through catalogs and physical stores. These behaviours make it a promising audience for loyalty initiatives and personalized product offers.

Recommended strategies:

- Introduce loyalty points, membership benefits, and rewards for repeat purchases.
- Offer relevant exclusive promotions and premium or complementary products.
- Provide personalized catalogs and recommendations based on purchase history.
- Encourage repeat purchases through targeted loyalty rewards.

Cluster 2 – Budget-conscious, independent customers

Customer profile: Lower income, a higher average number of children, and typically living alone.

Reported results: Average income: **$36,960** | Average spending: **$166** | Average children: **1.27** | Website visits: **6.66/month** | Campaign response: **14.19%** | Store purchases: **3.62**

Interpretation: This group has the lowest average income and spending. It also has the highest website-visit average. Its campaign response rate is higher than Cluster 0, which may provide an opportunity to convert browsing interest into purchases.

Recommended strategies:

- Offer personalized discounts and affordable product recommendations.
- Test limited-time promotions and small-value bundles of essential products.
- Use targeted digital campaigns with clear calls to action.
- Focus on converting website visits into completed purchases.

Cluster 3 – Premium, high-value independent customers

Customer profile: Higher income, fewer children on average, and typically living alone.

Reported results: Average income: **$70,723** | Average spending: **$1,190** | Average children: **0.46** | Catalog purchases: **5.01** | Store purchases: **8.43** | Campaign response: **32.01%**

Interpretation: This segment combines high average spending with the strongest campaign response rate. It represents an attractive audience for premium offers, personalized recommendations, and customer-retention activities.

Recommended strategies:

- Offer premium products, exclusive collections, and early access to selected launches.
- Test VIP memberships and personalized rewards.
- Recommend products based on customers' previous purchases.
- Develop retention initiatives to support long-term loyalty.

3. Comparison of all four clusters

The table summarizes the reported cluster averages and response rates. Spending is presented as average spending per customer, not total revenue generated by a cluster.

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Average income	$39,681	$72,808	$36,960	$70,723
Average spending	$222	$1,237	$166	$1,190
Average number of children	1.24	0.51	1.27	0.46
Website visits/month	6.31	3.58	6.66	3.73
Campaign response rate	7.62%	16.67%	14.19%	32.01%
Living arrangement	Partner	Partner	Alone	Alone

4. Overall business recommendations

The recommended priorities below are based on the combination of spending, campaign response, and engagement indicators. They represent suggested areas for testing and investment.

Priority	Cluster	Business objective
1	**Cluster 3**	Retain high-value customers and build on the segment's strong campaign response.
2	**Cluster 1**	Increase repeat purchases through loyalty programs and personalized offers.
3	**Cluster 2**	Convert frequent website visits into purchases using relevant digital promotions.
4	**Cluster 0**	Improve engagement through affordable, family-oriented offers and value promotions.

5. Key results and business insights

32.01%	$1,237	6.66	7.62%
Highest campaign response – Cluster 3	Highest average spending – Cluster 1	Most website visits per month – Cluster 2	Lowest campaign response – Cluster 0

- **Cluster 3** has the highest campaign response rate (32.01%), making it the strongest segment for testing targeted campaign strategies.
- **Cluster 1** has the highest average spending ($1,237), making it especially important for revenue-focused and loyalty initiatives.
- **Clusters 0 and 2** have higher website-visit averages but much lower spending, indicating a potential browsing-to-purchase conversion opportunity.

6. Additional analytical observation

The reported recency averages are relatively similar across all four clusters (approximately 48–50 days since the last purchase). This suggests that recency alone does not strongly distinguish the segments in this analysis. Income, spending, household composition, shopping behaviour, and campaign response provide more visible differences between the groups.

This does not mean recency is unimportant. It may still be useful for monitoring customer inactivity or identifying changes in purchasing patterns over time. However, it appears less useful as a standalone feature for explaining the four segments described here.

7. Key findings

- **Cluster 0:** Lower-spending customers with more children on average and the lowest campaign response. Affordable, family-oriented promotions may be worth testing.
- **Cluster 1:** Higher-income customers with the highest average spending and strong catalog/store purchasing activity. Loyalty and repeat-purchase strategies are a priority.
- **Cluster 2:** The lowest average income and spending, but the highest website-visit average. Digital conversion strategies may help turn browsing into purchases.
- **Cluster 3:** High-income, high-spending customers with the strongest campaign response. Premium offers and retention initiatives are promising areas to test.

8. Conclusion

The SmartCart clustering analysis demonstrates how customer data can be transformed into actionable business insights. The four segments show different spending levels, household characteristics, shopping patterns, and campaign responses. A single marketing strategy is unlikely to address all of these differences equally well.

SmartCart can use these segments to design more relevant campaigns: prioritize retention and premium engagement for Cluster 3, encourage loyalty and repeat purchases in Cluster 1, improve digital conversion in Cluster 2, and test affordable family-focused promotions in Cluster 0. Tracking campaign conversion, repeat purchase rate, average order value, and customer lifetime value will help determine which recommendations create measurable business impact.

9. Notes and interpretation limits

- The figures in this report are descriptive cluster averages and response rates supplied in the analysis. They describe the observed groups and do not establish that a marketing action will cause a particular result.
- Marketing recommendations are hypotheses to test. Their effectiveness should be evaluated through campaign experiments and business performance measures.
- The spending values are average customer spending figures, not the total revenue generated by each cluster.
- The segment labels (for example, “premium” or “budget-conscious”) are interpretive descriptions based on the reported averages, not labels assigned by the clustering algorithm.

Final takeaway: SmartCart's four customer segments provide a practical foundation for moving from one-size-fits-all marketing toward more personalized engagement, retention, and revenue strategies. The next step is to test the proposed campaigns and measure their actual impact.